

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

"By the end of these 6 sessions, you should be able to open any mutual fund fact sheet and know exactly which of these six questions its marketing is quietly avoiding. Which one would you guess — and why?"



Dr. (CMA) Srikanth Potharla, M.Com., M.Phil., FCMA., UGC-NET., Ph.D.

Assistant Professor, Department of Accounting & Finance · ICFAI Business School (IFHE Hyderabad)

Return Maximization vs Wealth Maximization: Why Averages Lie

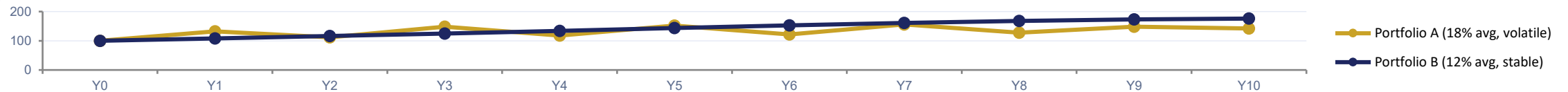
- Same 10-year horizon, two portfolios, two very different outcomes
- A higher average return does not guarantee higher terminal wealth
- The culprit: volatility compounds against you, not just against the average
- Return maximization $\neq$ wealth maximization — the throughline of this entire session

PORTFOLIO A

18% average annual return

Volatility: Very High

₹100 grows to ₹142 over 10 years



PORTFOLIO B

12% average annual return

Volatility: Stable

₹100 grows to ₹176 over 10 years

*“Should investors pursue the highest average return, or the most efficient wealth-creation process?
Why do fund advertisements almost always choose to report the former?”*

QUESTION BANK LINK

No direct problem for this framing slide — it foreshadows Problems 1.2 (arithmetic mean) and 1.3 (geometric mean), used next.

Single-Period Return & Arithmetic Mean: A Forecasting Tool, Not a Report Card

- Single-period return: $R = (\text{Ending Value} - \text{Beginning Value} + \text{Income}) / \text{Beginning Value}$
- Arithmetic mean: simple average of periodic returns — $R = \sum R_i / n$
- Best use: estimating expected return for a single future period
- Weakest use: describing what actually happened to accumulated wealth over many periods

Year	Y1	Y2	Y3	Y4	Y5
Return	+20%	+15%	+10%	−5%	+10%



ARITHMETIC MEAN

10%

Forecasting tool — not a performance report card.

Five illustrative annual returns, averaged to estimate one forward-looking period.

PROFESSOR CUE

“SEBI/AMFI fact sheets report arithmetic-style returns. If arithmetic mean is best suited for forecasting a single future period, what exactly are investors assuming when they read it as a track record?”

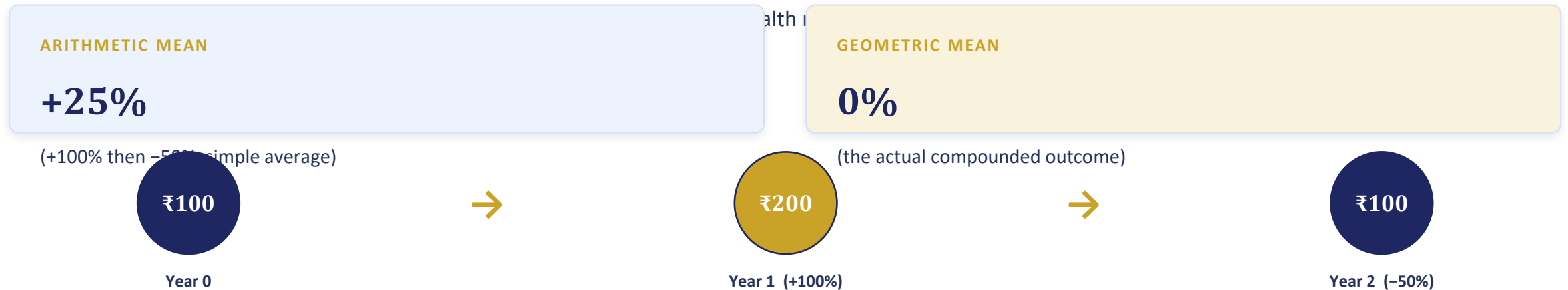
QUESTION BANK LINK

Use in class: Problem 1.1 (single-period return)

Assign for practice: Problem 1.2 (arithmetic mean return)

Geometric Mean Return: The Compounding Reality Behind Reported Gains

- Geometric mean: the true compound annual growth rate of wealth
- $GM = [\prod(1+R_i)]^{1/n} - 1$ (formula stated, not derived in class)
- Volatility drag: the wider the swings, the larger the arithmetic–geometric gap



Same two returns, opposite conclusions depending on which mean you ask.

PROFESSOR CUE

“Should SEBI or AMFI mandate geometric mean disclosure for equity funds with a track record beyond three years? Would this change how retail investors select funds?”

QUESTION BANK LINK

Use in class: Problem 1.3 (geometric mean return)

Holding Period Return & Effective Annual Return: Why We Annualize

- HPR: total return over any investment horizon, regardless of length
- EAR: HPR converted to a standardized annual-equivalent basis
- HPR = EAR only when the holding period is exactly one year
- Annualizing makes a 6-month FD, a 1-year bond, and a 25-year SIP comparable on equal footing

6-MONTH HOLDING

HPR = 8%

EAR $\approx$ 16.6%

Short horizon — compounding effect is small but already visible

1-YEAR HOLDING

HPR = 15%

EAR = 15%

HPR = EAR exactly at the one-year mark

25-YEAR HOLDING

HPR $\approx$ 2,000%*

EAR $\approx$ 13.0%

A huge cumulative number, a modest annual rate

**Figures illustrative for teaching — exact values worked in Question Bank Problems 2.1–2.3.*

PROFESSOR CUE

“A 25-year investment might show a 2,000% cumulative HPR — an impressive-sounding number. What does the EAR reveal that the headline HPR conceals?”

QUESTION BANK LINK

Use in class: Problem 2.1 (six-month HPR/EAR)

Assign for practice: Problems 2.2 (one-year), 2.3 (twenty-five-year)

Variance & Standard Deviation: Quantifying Total Risk

- Variance: average squared deviation of returns from the mean
- Standard deviation (σ): square root of variance — same units as returns, interpretable
- Every downstream model (CAPM, Sharpe ratio, tracking error, VaR) is built on SD, not variance
- Population variance ($\div n$) for description vs sample variance ($\div n-1$) for forward estimation — know which you're using

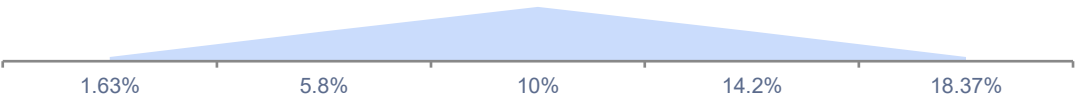
Year	Return	Deviation	Squared Dev.
Y1	20%	+10	100
Y2	15%	+5	25
Y3	10%	0	0
Y4	5%	-5	25
Y5	0%	-10	100

Illustrative deviation table — mean return = 10%

$\sigma^2 = 70 \rightarrow \sigma = 8.37\%$

Same units as returns — directly interpretable, unlike variance.

$\pm 1\sigma$ band around a 10% mean



PROFESSOR CUE

“Does higher variance always mean a worse investment? Consider a positively skewed, high-variance asset — is all volatility equally undesirable?”

QUESTION BANK LINK

Use in class: Problem 1.4 (standard deviation of returns)

Assign for practice: Problem 4.2 (variance/SD of market and security)

Coefficient of Variation: Comparing Risk Across Dissimilar Assets

- Coefficient of Variation: $CV = \sigma / \text{Mean Return}$ — risk per unit of return
- Lower CV = more efficient risk-taking, regardless of absolute standard deviation
- Limitation: not meaningfully interpretable when mean return is at or near zero (Slide 9)

Metric	Investment A	Investment B	Better?
Expected Return	7%	12%	
Standard Deviation	5%	7%	
Coefficient of Variation	0.71	0.58	
Lower CV — more efficient		✓ Investment B	

PROFESSOR CUE

“Should investors ever evaluate risk in isolation from return? When, if ever, might standard deviation alone be sufficient?”

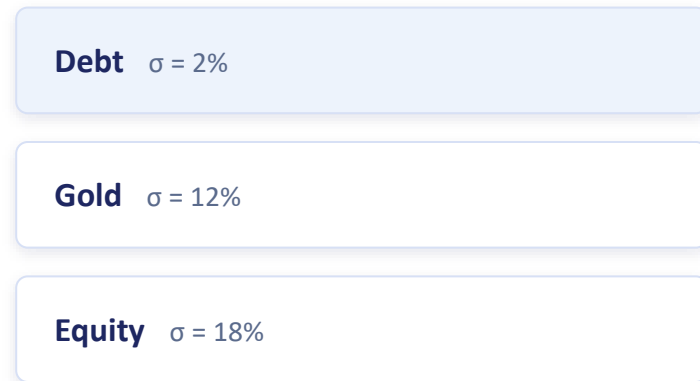
QUESTION BANK LINK

Use in class: Problem 9.1 (Fund P vs Fund Q coefficient of variation comparison)

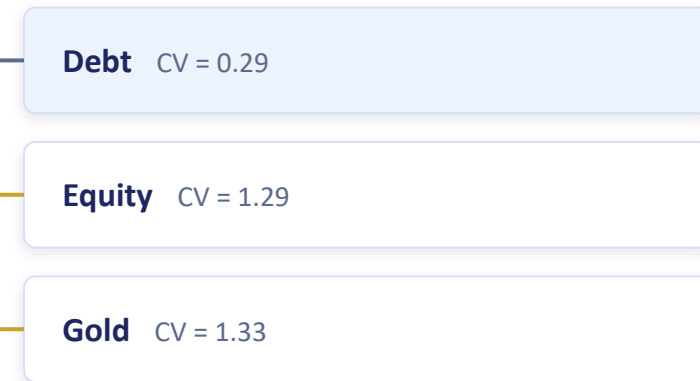
Coefficient of Variation: Ranking Reversals Across Asset Classes

- Same CV logic, now applied across genuinely different asset classes
- Debt: 7% return, $\sigma=2\%$ | Equity: 14% return, $\sigma=18\%$ | Gold: 9% return, $\sigma=12\%$
- Raw SD ranking: Debt < Gold < Equity — CV ranking: Debt < Equity < Gold

BY RAW STANDARD DEVIATION



BY COEFFICIENT OF VARIATION



Gold and Equity swap positions once return is factored in — Debt stays most efficient either way.

PROFESSOR CUE

“A wealth manager pitches gold as ‘the safer diversifier’ because its standalone SD is lower than equity’s. Does the coefficient of variation support that pitch?”

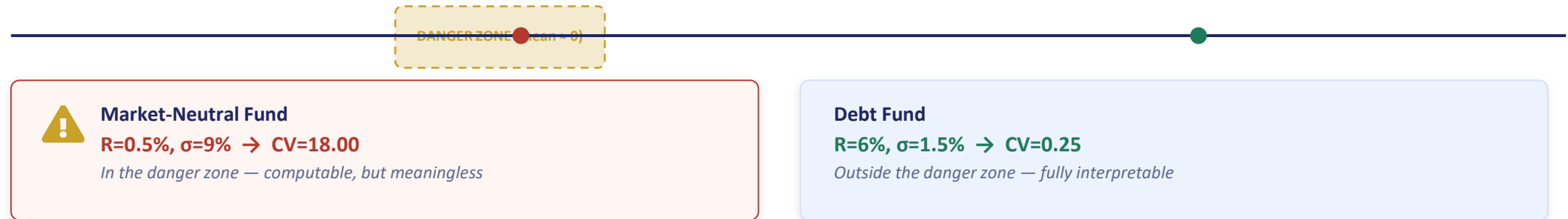
QUESTION BANK LINK

Assign for practice: Problem 9.2 (three-asset-class CV ranking reversal — debt, equity, gold)

When Coefficient of Variation Breaks Down: The Near-Zero-Return Trap

- CV is a ratio — ratios misbehave when the denominator approaches zero
- Market-neutral fund: $R=0.5\%$, $\sigma=9\%$ → $CV = 18.00$ (technically computable, practically meaningless)
- Debt fund: $R=6\%$, $\sigma=1.5\%$ → $CV = 0.25$ (fully interpretable)

MEAN RETURN →



PROFESSOR CUE

“If a metric can be calculated but not meaningfully interpreted, should it still be reported on a fund fact sheet? Who decides?”

QUESTION BANK LINK

Problem 9.3 (CV instability near a zero-mean return)

What Exactly Is Risk? Five Practitioners, Five Definitions

- Risk means something different depending on who is asking — and why



Retail Investor

Loss of capital I can't afford to lose



Equity Fund Manager

*Underperforming the benchmark
(career risk)*



Pension Fund Manager

*Liability mismatch — decades-out
shortfall*



Wealth Manager

*Volatility the client can't
behaviourally tolerate*



Behavioural Researcher

*Probability-weighted regret —
losses hurt $\approx 2\times$ gains*

PROFESSOR CUE

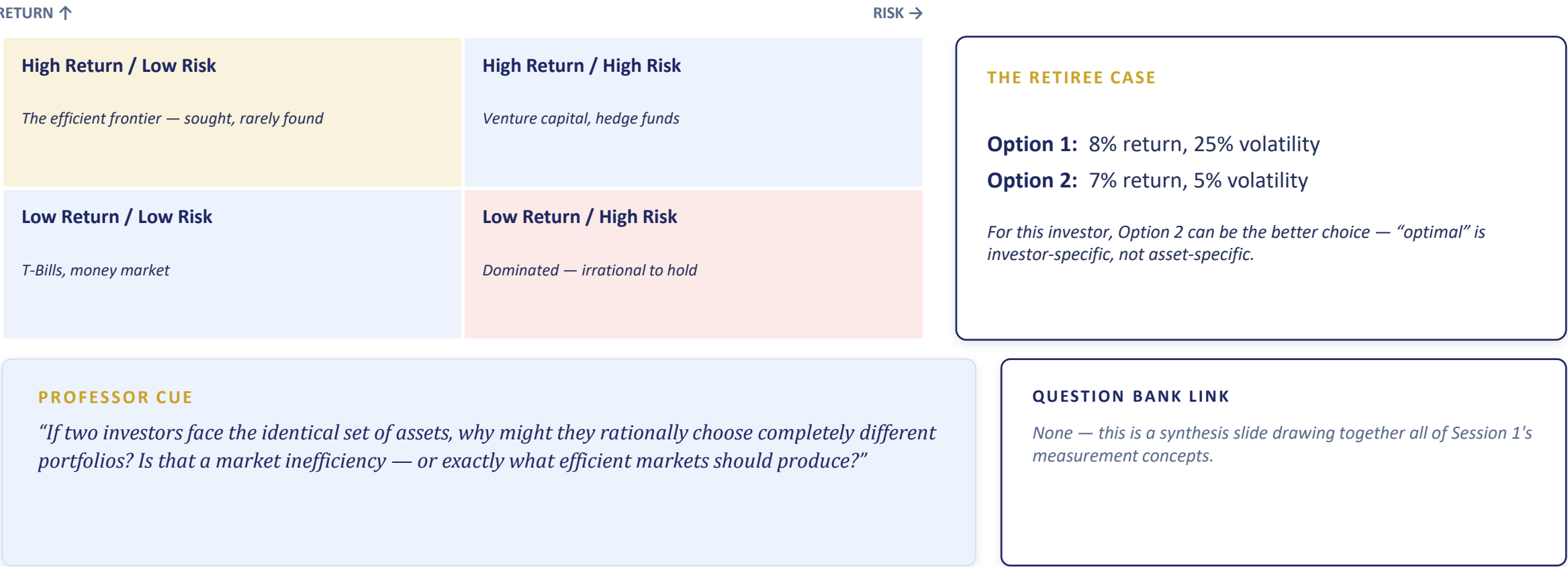
“Is volatility truly risk, or merely uncertainty? Does your answer change with investment horizon and investor type?”

QUESTION BANK LINK

None — conceptual bridge slide. Sets up the utility-based optimization discussion in Slide 11.

The Risk-Return Trade-off: What Are Investors Really Optimizing?

- Return maximization is meaningless without risk context
- Utility theory: investors assign different value to the same return based on risk tolerance
- Institutional constraints (pension funds, endowments, insurers) redefine “optimal” through liability-matching



Session 1 Synthesis: The Measurement Foundations Before Portfolio Theory

- Risk has no single definition — it depends on who is asking and why
- Return maximization without risk context is not optimization — it's guessing

RETURN MEASURES BUILT

Single-Period Return → Arithmetic Mean → Geometric Mean →
HPR → EAR

RISK MEASURES BUILT

Variance → Standard Deviation → Coefficient of Variation

?

Every measure so far describes ONE asset in isolation. Session 2 → Covariance, Correlation, Beta & Alpha

PROFESSOR CUE

"If return and risk are both fully measured for every asset individually, what problem still remains completely unsolved?"

QUESTION BANK LINK

None — pure recap/bridge slide closing Session 1.